

Seán Varian

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Professional Summary

Electronics engineer with 3 years of industry experience as an Analog Design Engineer at u-blox, complemented by extensive hands-on project work in robotics, embedded systems, CMOS design and prototyping. Skilled in circuit design, PCB layout, lab validation, and mechatronic system integration. Proficient with industry tools such as Cadence, SolidWorks and OnShape. I also have experience working with lab equipment such as oscilloscopes, spectrum/network analyzers, function generators etc. Also, I have experience working in large scale on-site, industrial settings where I have exposure to PLC's, 3-phase power systems, motor drivers, VFD's etc. I am passionate about developing innovative, practical solutions that merge electronics and mechanics. Seeking a hands-on role in electronics design, mechatronics, R&D, validation, or commissioning.

Key Skills

- **Analog & RF:** Circuit design, simulation, debugging, and layout (Cadence)
- **Embedded systems:** Arduino, C++, Python (PCEP certified)
- **Mechatronics prototyping:** 3D printing, SolidWorks, OnShape
- **Mechanics:** Mechanical design, tolerancing, assemblies, enclosure design
- **PLC programming:** Ladder logic, structured text, CodeSys
- **PCB Fabrication:** SMD/THT soldering, PCB assembly & rework
- **Lab validation:** oscilloscopes, DMMs, spectrum/network analyzers
- **Documentation:** Technical report writing & competency with microsoft word and excel

Selected Projects

- **Ultrasonic Particle Levitation System** – Designed and built a 72-transducer levitation system (based on Bristol's TinyLev) controlled via Arduino. Demonstrated levitation and manipulation of both solid and liquid particles for my college final year project. This project involved the design of the mechanical structure and alignment system for the transducer array using CAD.
- **Engine Control Unit (ECU)** – Developed an ECU using a microcontroller and sensors/actuators to measure, store, and process engine parameters.
- **Obstacle-Avoiding Robot Car** – Designed a self-driving robotic car using Arduino Uno, ultrasonic sensors, and motor drivers. This project involved the design of a mechanical

chassis which the electronics were mounted on. The robotic car can detect surrounding obstacles and navigate around them.

- **IoT Self-Watering Plant System** – Built a moisture/temperature-sensing irrigation system with cloud integration and 5V pump control using Arduino's IoT platform to monitor the system's parameters and performance.
- **Automated Light Switch** – Designed a rack-and-pinion mechanism with servo control and ESP32-IR communication for remote mechanical switching of a light switch across a room. This system detects light using photoresistors and uses a servo to switch the light on or off depending on the light in the room.

Professional Experience

Commissioning Engineer

Liebherr Container Cranes | Killarney, Ireland | 11/2025 – present

- I work as part of the commissioning team where I travel to ports around the world ensuring that the container cranes operate properly, safely and efficiently according to the design specifications.
- Worked on large-scale electromechanical systems including motors, drive systems and crane motion mechanisms
- Oversee crane operation, both STS and RTG, from initial installation to customer handover.
- Perform troubleshooting and debugging of PLC ladder logic and structured text using Codesys.
- Carry out electrical repair work within electrical panels, specifically on PLC and drive modules.
- Conduct thorough technical documentation and reports on each project

Analog IC Design Engineer

U-blox | Cork, Ireland | 06/2023 – 06/2025

- Designed, simulated, and validated analog building blocks (LDOs, ESD clamps, PFD for PLLs).
- Performed IC layout and supported first-pass silicon success.
- Conducted lab validation and characterization for GNSS-related chips and modules.
- Created technical documentation to communicate engineering specifications to production teams.

Hardware Engineering Intern

Dell Technologies | Cork, Ireland | 05/2022 – 09/2022

- Designed and ran validation tests for solid-state and hard-disk drives.
- Developed Python scripts for a new Linux-based testing environment.

- Coordinated server rack installations and optimized cable lengths for improved power efficiency.

Electrical Engineering Intern

Jacobs Engineering | Cork, Ireland | 05/2021 – 09/2021

- Contributed to electrical design for a major Pfizer plant expansion.
- Performed load calculations, cable sizing, and circuit protection design.
- Maintained documentation of electrical design changes for compliance.

Education

BEng Electrical & Electronic Engineering

University College Cork | 09/2019 – 05/2023

- Final Year Project: Design and Test of an Ultrasonic Particle Levitation System
- Modules: Digital/Analog IC design, DSP, RF design, Machine Learning, Embedded Systems, Photonics, Telecommunications, Mechanics.

Languages

English: Fluent

Irish (Gaelic): Conversational